

Lab 05a

2D Array

1 Review

- 2D arrays are used to store data in a table or grid format, where each element can be accessed using two indices (row and column).

- + They are particularly useful for representing matrices, tables, or any other data structure that requires a grid-like organization.

- + You can declare, initialize, and manipulate these arrays using nested loops and standard 1D array operations.

2 Requirements

2.1 In-class assignment

2.1.1. P315

Write a function to find the largest value in a matrix of real numbers.

2.1.2. P320

Calculate the sum of positive numbers in a matrix of real numbers.

2.2 Homework

2.2.1. P322

Calculate the sum of the values in each row of a matrix of real numbers.

2.2.2. P347

Count the number of "Saddle Points" in the matrix. An element is called a Saddle Point if it is the largest in its row and the smallest in its column.

2.2.3. P398

Rotate the values on the border of the matrix clockwise.

2.2.4. P423

Sort the values in the matrix in ascending order in a spiral pattern (spiral matrix).

2.2.5. P457

Check if the matrix is symmetric with respect to the main diagonal

2.2.6. P471

Calculate the product of two matrices.

3 Guidelines

Create your project with appropriate .h/.cpp files.

Write at least 1 function for each problem.

Write a main() function, which a menu that allows user to choose a problem to be executed If that user chooses 0, exit the program

```
// <YOUR STUDENT ID>  
// <YOUR FULL NAME>  
// <YOUR CLASS>
```

```
// Test case 1
```

```
// Input:
```

```
// Output:
```

```
...
```

```
// Test case 3
```

```
// Input:
```

```
// Output:
```

```
int main(){  
    //your code here  
  
    return 0;  
}
```